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Appendix

A Proofs

Proof of Theorem 3.3. Suppose λ and λ' are two tasks with primitive decompositions

$$\lambda' = \lambda'_1 \circ \cdots \circ \lambda'_{T'}$$

and

$$\lambda = \lambda_1 \circ \cdots \circ \lambda_T, \tag{8}$$

where $\arg\min_t\{\lambda_t \notin \mathrm{Span}(\{\lambda'_1, \ldots, \lambda'_{T'}\})\} \leq k$. In other words, the primitive decompositions of λ' and λ diverge before step $k+1$. Then, Assumption 3.1 implies that for any $j \geq k$, we have that the answer Y and X'_j are d-separated by X'_{j-1}. Therefore,

$$Y \perp\!\!\!\perp X'_j \mid X'_{j-1}.$$

Next, we know from Assumption 3.2 that there exists some task $\hat{\lambda} \in \mathrm{Span}(\Gamma^M)$ (possibly dependent on X_0 and λ) such that $\Lambda^M(X_0, \lambda) \stackrel{\mathrm{d}}{=} \Lambda(X_0, \hat{\lambda})$. Suppose that $\hat{\lambda}$ has primitive decomposition

$$\hat{\lambda} = \tilde{\lambda}_1 \circ \cdots \circ \tilde{\lambda}_{\tilde{T}},$$

then since $\hat{\lambda} \in \mathrm{Span}(\Gamma^M)$, we know that $\tilde{\lambda}_i \in \Gamma^M$ for $i \in \{1, \ldots, \tilde{T}\}$. If the primitive decomposition of λ in (8) is such that $k = \arg\min_t\{\lambda_t \notin \mathrm{Span}(\Gamma^M)\}$, then we know that $\arg\min_t\{\lambda_t \notin \mathrm{Span}(\{\tilde{\lambda}_1, \ldots, \tilde{\lambda}_{\tilde{T}}\})\} \leq k$. Then, from the above it follows that

$$Y \perp\!\!\!\perp X^M_j \mid X^M_{j-1}.$$

Here, we used the fact that $X^M_j \stackrel{\mathrm{d}}{=} \Lambda(X_0, \tilde{\lambda}_1 \circ \cdots \circ \tilde{\lambda}_j)$ using Assumption 3.2.

□

Proof of Proposition 3.4.

$$\begin{aligned}
\mathbb{E}[\log p(Y \mid X^M_j)] - \mathbb{E}[\log p(Y \mid X^M_{j-1})] &= \mathbb{E}\left[\log \frac{p(Y \mid X^M_j)}{p(Y \mid X^M_{j-1})}\right] \\
&= \mathbb{E}\left[\log \frac{p(Y \mid X^M_j, X^M_{j-1})}{p(Y \mid X^M_{j-1})}\right] \\
&= \mathbb{E}\left[\log \frac{p(Y, X^M_j \mid X^M_{j-1})}{p(Y \mid X^M_{j-1})\, p(X^M_j \mid X^M_{j-1})}\right] \\
&= \mathcal{I}(Y, X^M_j \mid X^M_{j-1})
\end{aligned} \tag{9}$$

Here, the second equality above arises from the fact that X^M_j also captures all the information captured in X^M_{j-1} (and possibly more). Therefore, conditional on X^M_j, the state X^M_{j-1} is deterministic and hence, $Y \perp\!\!\!\perp X^M_{j-1} \mid X^M_j$. □

A.1 Symmetry property of information-gain $\mathcal{I}(Y, X^M_j \mid X^M_{j-1})$

The mutual information between two random variables X and Y, $\mathcal{I}(X, Y)$, is symmetric in its arguments (i.e., w.r.t. X and Y). However, the conditional mutual information $\mathcal{I}(Y, X^M_j \mid X^M_{j-1})$ satisfies a symmetry property <u>conditional</u> on X^M_{j-1}. Formally, this property of the information-gain term can be expressed as follows:

There exists some functional $\mathcal{F} : \mathcal{P} \times \mathcal{P} \times \mathcal{P} \to \mathbb{R}$ where $\mathcal{P}$ is the space of probability distributions, such that

1. The information-gain $\mathcal{I}(Y, X_j^M \mid X_{j-1}^M)$ can be expressed as:

$$\mathcal{I}(Y, X_j^M \mid X_{j-1}^M) = E\left[\mathcal{F}\left(P_{Y|X_{j-1}^M}, P_{X_j^M|X_{j-1}^M}, P_{Y,X_j^M|X_{j-1}^M}\right)\right],$$

   where P_Z denotes the distribution of the random variable Z and

2. $\mathcal{F}$ is symmetric w.r.t. its first two arguments, i.e. $\mathcal{F}(p, q, r) = \mathcal{F}(q, p, r)$. Note that there is no symmetry requirement w.r.t. the third argument of $\mathcal{F}$ because the joint distribution $P_{Y,X_j^M|X_{j-1}^M}$ is already symmetric w.r.t. Y and X_j^M i.e. $P_{Y,X_j^M|X_{j-1}^M} = P_{X_j^M,Y|X_{j-1}^M}$.

It follows from Eq. (9) in the proof of Proposition 3.4, that the functional $\mathcal{F}$ which satisfies the above conditions is

$$\mathcal{F}(p, q, r) = -E_{X\sim p}[\log p(X)] - E_{X\sim q}[\log q(X)] + E_{X\sim r}[\log r(X)].$$

B Additional details of our framework

B.1 State-conditioned unidentifiability

In practice, the concept of unidentifiability may depend on the initial state X_0. For instance, an LLM might accurately perform addition for 2-digit numbers but fail with 10-digit numbers [24]. Our framework can be extended to account for such cases by explicitly incorporating the distribution of initial states into the notion of identifiability. For example, addition could be considered unidentifiable when the initial state distribution is $p(X_0 \mid X_0$ includes 10-digit numbers$)$. However, for simplicity, we keep this distributional dependence implicit in our framework.

B.2 How to define steps beyond mathematical datasets

The framework presented in this paper primarily considers examples related to mathematical reasoning, where the definition of primitive tasks is intuitive and well-structured. However, our methodology could be applied to other domains where the identification of primitive tasks is less straightforward, such as Blocks World [25, 33] and commonsense question answering (QA) [26].

In the case of Blocks World, the primary task of planning can be decomposed into sub-tasks involving sequences of primitive actions, such as “stack", “unstack" and “move". Applying the information-gain methodology in this context could provide insights into the effectiveness of large language models (LLMs) in planning and executing these sub-tasks. By analyzing the information-gain for each step, it would be possible to assess where the model’s reasoning process is effective and where it encounters difficulties.

Similarly, in commonsense QA, the chain-of-thought (CoT) reasoning steps can be categorized into distinct types, such as causal reasoning (identifying cause-and-effect relationships), temporal reasoning (understanding sequences and timing), and spatial reasoning (comprehending physical arrangements and object relationships). These reasoning types align naturally with the proposed framework, enabling a systematic evaluation of the LLM’s decision-making process within each category.

While these extensions present promising directions for future research, they also introduce additional challenges, particularly regarding the assumptions underlying our methodology. Addressing these challenges and validating the framework across diverse domains remains an open avenue for further investigation.

B.3 Correct final answers with incorrect intermediate steps

A notable phenomenon of chain-of-thought (CoT) reasoning is the occurrence of cases where a model arrives at the correct final answer despite containing errors in intermediate steps. This scenario raises important questions regarding the validity of intermediate reasoning and the implications for evaluating model performance.

The methodology proposed in this paper estimates the information contributed by each successive reasoning step toward the final correct answer. In instances where an intermediate step is incorrect, it is expected that this step contributes no additional relevant information, resulting in an information-gain of zero at that